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On odd colorings of planar graphs ☆

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ABSTRACT

An *odd k-coloring* of a graph G is a proper k-coloring such that every non-isolated vertex in G has a color appearing odd times on its neighborhood. In 2022, Petruševski and Škrekovski [Colorings with neighborhood parity condition, Discrete Appl. Math., 321: 385-391] introduced this concept and conjectured that every planar graph admits an odd 5-coloring. We prove that (1) every planar graph without adjacent 3-cycles is odd 7-colorable; (2) If G is a planar graph without 3-cycles and intersecting 4-cycles, then G is odd 5-colorable; (3) The odd chromatic number of a Halin graph is either 3 or 4.
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1. Introduction

Let G be a graph. A (vertex) coloring of G is proper if every two adjacent vertices receive distinct colors. Denote the *(vertex) chromatic number* of G by $\chi(G)$. In 1941, Brooks stated that if G is a connected graph, and is neither an odd cycle nor a complete graph, then $\chi(G) \leq \Delta(G)$. The proper coloring on planar graphs originated from the well known Four-Color Conjecture, which states that every loopless planar graph is 4-colorable. Recently, two groups of scholars introduced the concepts of conflict-free coloring of hypergraphs [15] and graphs [9], respectively. A *conflict-free k-coloring* of G is a coloring $\phi : V \to \{1, 2, \ldots, k\}$ such that for every non-isolated vertex $v \in V$, there exists a color on the neighborhood of v appearing exactly once. The smallest k such that G admits a conflict-free k-coloring (CF k-coloring, for short) is the *CF chromatic number* of G, denoted by $\chi_{cf}(G)$. Assume G is a graph with n vertices. In 2009, Cheilaris [9] proved that the upper bound of the CF chromatic number of G is $2\sqrt{n}$. Later, Pach and Tardos [20] improved it to $O(\log^{2+\varepsilon} n)$ for any $\varepsilon > 0$. Planar graphs and minor-k-colorable graphs [18] were also investigated. The conflict-free variant of the famous Hadwiger Conjecture [17] was verified in [1]. For more information on CF colorings of hypergraphs and graphs, we refer to readers to [5,24,3,4,6].

Combining proper coloring and CF coloring, Fabrici, Lužar, Rindošová, and Soták [16] introduced the proper conflict-free coloring. If a CF k-coloring is proper, then we call it a *proper conflict-free (PCF) k-coloring*. The *PCF chromatic number* of a graph G is denoted by $\chi_{pcf}(G)$. It is NP-complete to determine the PCF chromatic number of a graph [8], as well as determining whether a bipartite (planar, respectively) graph admits a PCF k-coloring for $k \geq 3$ ($k = 4$, respectively) [2].

A natural variant of the PCF coloring of graphs was independently introduced in 2013 [10] and 2022 [22]. Given a graph G, an *odd coloring* of G is a proper coloring such that for every non-isolated vertex v in G, there exists a color that occurs odd times on the neighborhood of v. A graph is *odd k-colorable* if it admits an odd k-coloring. The *odd chromatic*

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number of G, denoted by $\chi_o(G)$, is the smallest integer k such that G admits an odd k-coloring. Clearly, it holds that $\chi(G) \le \chi_o(G) \le \chi_{pcf}(G)$ for any graph G. Similar to PCF coloring, determining the odd chromatic number of a graph [7] and whether a bipartite graph is odd k-colorable for $k \ge 3$ [2] are two NP-complete problems. In this paper, we mainly focus on odd colorings of graphs.

Cranston studied the odd coloring of graphs with bounded maximum average degree. The *maximum average degree* of graph G, denoted by *mad(G)*, is equal to $\max\{\frac{2|E(H)|}{|V(H)|} \mid H \subseteq G\}$. Cranston [13] proved that for every ϵ with $0 < \epsilon \le \frac{8}{5}$, if $mad(G) \le 4-\epsilon$, then $\chi_o(G) \le \lfloor \frac{8}{\epsilon} \rfloor + 2$. Moveover, as $\epsilon \to 0$, infinitely often there exists a graph G_ϵ such that $mad(G_\epsilon) = 4-\epsilon$ and $\chi_o(G_\epsilon) = \lfloor \frac{8}{\epsilon} \rfloor - 1$. Based on this, the following conjecture was proposed.

Conjecture 1.1 *([13]). Let G be a graph. For every ϵ with $0 < \epsilon \le \frac{8}{5}$, if $mad(G) \le 4-\epsilon$, then $\chi_o(G) \le \lfloor \frac{8}{\epsilon} \rfloor - 1$.*

In the same paper, Cranston pointed out that for some fixed ϵ, the upper bound of odd chromatic number might be better than in Conjecture 1.1. For example, if $\epsilon = 1$, then $\chi_o(G) \le 6$; If $\epsilon = \frac{8}{7}$, then $\chi_o(G) \le 5$. Recently, Cho, Choi, Kwon, and Park [12] verified that Conjecture 1.1 holds for $0 < \epsilon < \frac{8}{5}$. However, Conjecture 1.1 does not hold when $\epsilon = \frac{8}{5}$. Indeed, it holds that $mad(C_5) = 2 < 4 - 8/5$, but $\chi_o(C_5) = 5 > \lfloor \frac{8}{8/5} \rfloor - 1$. With this counterexample, they proved that every graph G with $mad(G) \le \frac{22}{9}$ and without induced 5-cycle is odd 4-colorable.

Actually, when the odd coloring was introduced, Petruševski and Škrekovski [22] studied planar graphs. As $\chi_o(C_5) = 5$ and the fact that $\chi_o(G) \ge \chi(G)$ for any graph G, they proposed the following conjecture.

Conjecture 1.2 *([22]). Every planar graph admits an odd 5-coloring.*

Started from Petruševski and Škrekovski [22], they verified that every planar graph is odd 9-colorable. Later, Petr and Portier [21] improved this upper bound to 8. Motivated from Conjecture 1.1, we know intuitively that sparser graphs have smaller odd chromatic number. Tian and Yin [25] showed that every toroidal graph with girth at least 4 is odd 7-colorable. Cho, Choi, Kwon, and Park studied planar graphs with restriction on girth. They pointed out that [12] a sufficient condition for G to be odd 6-colorable is that G is a planar graph with girth at least 5.

Two cycles are said to be *adjacent* if they share a common edge and are said to be *intersecting* if they share a common vertex. A cycle of length k is called a *k-cycle*. Let G be a planar graph without intersecting 3-cycles. We claim that G admits an odd 6-coloring. Indeed, G can be vertex-partitioned to two forests (see [11], Theorem 2), denoted by T_1 and T_2. Since every tree is odd 3-colorable [13], we infer that $\chi_o(G) \le \chi_o(T_1) + \chi_o(T_2) = 6$. In this paper, we support Conjecture 1.2 by proving Theorem 1.3 and Theorem 1.4, which improve the main results in [12] and [25].

Theorem 1.3. *Every planar graph without adjacent 3-cycles admits an odd 7-coloring.*

Theorem 1.4. *If G is a planar graph without 3-cycles and intersecting 4-cycles, then G admits an odd 5-coloring.*

Besides planar graphs with bounded girth, some special embedded graphs were investigated, such as 1-planar graphs [14,19], outer planar graphs [7], outer-1-planar graphs and 2-boundary planar graphs [23]. In this paper, we study a class of special planar graphs, Halin graphs. A *Halin graph* is a planar graph $H = T \cup C$, where T is a tree embedded in a plane with at least four vertices and contains no vertex of degree two, and C is a cycle connecting the leaves of T consecutively in circular order. We show tight lower and upper bounds of odd chromatic number of Halin graphs in Theorem 1.5.

Theorem 1.5. *If G is a Halin graph, then $3 \le \chi_o(H) \le 4$. Moveover, these two bounds are tight.*

We organize this paper as follows. The proofs of Theorem 1.3, Theorem 1.4 and Theorem 1.5 will be given in Section 3, Section 4 and Section 5, respectively. Before proving the theorems, we introduce some notations, definitions and observations in Section 2.

2. Preliminaries

We use the convention "$S :=$" to mean that S is defined to be the right-hand side of the relation. For any two integers a, b with $a \le b$, we denote the set $\{i \in \mathbb{Z} \mid a \le i \le b\}$ by $[a, b]$ for short.

Assume G is a graph and u is an arbitrary vertex in G. We use $N_G(u)$ to denote the *neighborhood* of u in G. Define $d_G(u) := |N_G(u)|$ the *degree* of u in G and $\delta(G) := \min\{d_G(u) \mid u \in V(G)\}$ the *minimum degree* of G. If $d_G(u) = d$ ($d_G(u) \ge d$, $d_G(u) \le d$, respectively), u is called a *d-vertex* (*d^+-vertex*, *d^--vertex*, respectively) in G. We call u an *odd-vertex* in G if $d_G(u)$ is odd and call u an *even-vertex* in G otherwise. Define $V_{d^+}(G) := \{u \mid d_G(u) \ge d, u \in V(G)\}$ and $V_{d^-}(G) := \{u \mid d_G(u) \le d, u \in V(G)\}$. Assume uv is an edge in G and $d_G(v) = k$, v is called a *k-neighbor* of u. We call v an *odd-neighbor* of u if k is odd and call v an *even-neighbor* of u otherwise.

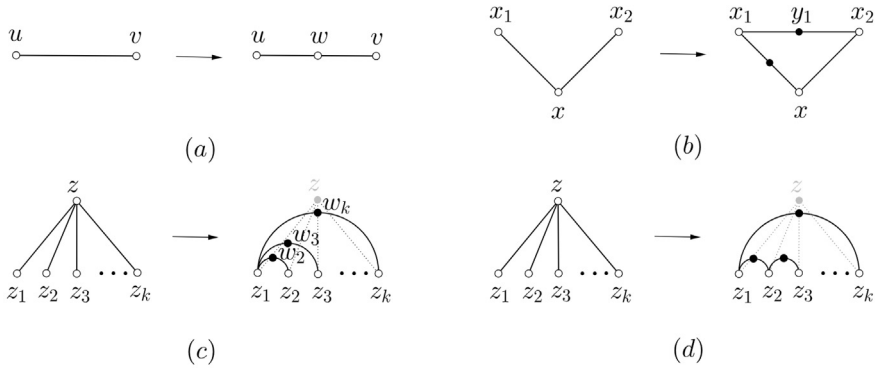

Fig. 1. Illustration of (A1)-(A4): dotted lines and gray points indicate the removed edges and vertices, respectively.

Let φ be an arbitrary odd coloring of G. We call the color appearing odd times on $N_G(u)$ the *odd color* of u. Note that every vertex of degree two in G has exactly two odd colors under φ. In addition, an odd-vertex in G always has an odd color under an arbitrary vertex coloring of G. As every vertex u has an odd color under φ, we choose an arbitrary odd color of u and denote it by $\varphi_o(u)$. For any $V \subseteq V(G)$, define that

$$\varphi(V) := \{\varphi(v) \mid v \in V\}, \qquad \varphi_o(V) := \{\varphi_o(v) \mid v \in V\}.$$

Let ϕ be a vertex coloring of G. Note that if every vertex in V has no odd color, then $\phi_o(V) = \emptyset$.

In the proofs of Theorem 1.3 and Theorem 1.4, we will construct some auxiliary graphs that are sparser than their corresponding original graphs. Let $\mu(G) = \frac{|V_{3+}(G)|}{|V(G)|}$ for any graph G. Given two graphs G_1, G_2, if $\mu(G_1) < \mu(G_2)$, then we say G_1 is *sparser* than G_2, denoted by $G_1 \prec G_2$. Let G be a planar graph. We construct some auxiliary graphs of G that are sparser than G by increasing the number of 2-vertices or decreasing the number of 3^+-vertices referring to $(A1) - (A4)$. Let uv be an edge in G. The *subdivision* of uv is to replace uv by a path uwv of length two, where w is a new 2-vertex. The auxiliary graphs are defined as follows.

(A1) Assume uv is an edge in G. Define G_1 the graph obtained from G by subdividing uv, as shown in Fig. 1(a).

(A2) Let x be a 2-vertex in G and let x_1, x_2 be the two neighbors of x. Assume that x_1, x_2 are 3^+-vertices. Define G_2 the graph obtained from G by subdividing xx_1 and
- if x_1 and x_2 have a common 2-neighbor different from x in G, denote it by y_1;
- otherwise, join x_1 and x_2 with a path denoted by $x_1y_1x_2$, as shown in Fig. 1(b).

Assume z is an arbitrary k-vertex in G and $N_G(z) = \{z_1, z_2, \ldots, z_k\}$, where $k \geq 3$ and z_i is a 3^+-vertex for every $i \in [1, k]$.

(A3) Define G_3 the graph obtained from G by removing the vertex z and joining z_1 and z_i with a path denoted by $z_1w_iz_i$ for each $i \in [2, k]$, as shown in Fig. 1(c).

(A4) Define G_4 the graph obtained from G by removing the vertex z and joining z_k and z_1, z_i and z_{i+1} with paths of length two for every $i \in [1, k-1]$, as shown in Fig. 1(d).

From (A1)-(A4), the following two observations are true obviously.

Observation 1. *Let G be a planar graph without adjacent 3-cycles. For $i \in [1, 4]$, if G_i is a graph obtained from G by applying (Ai), then*

(1) *G_i is a planar graph without adjacent 3-cycles; and*
(2) *$G_i \prec G$.*

Observation 2. *Let G be a planar graph without 3-cycles and intersecting 4-cycles. For $i \in [1, 2]$, if G_i is a graph obtained from G by applying (Ai), then*

(1) *G_i is a planar graph without 3-cycles and intersecting 4-cycles; and*
(2) *$G_i \prec G$.*

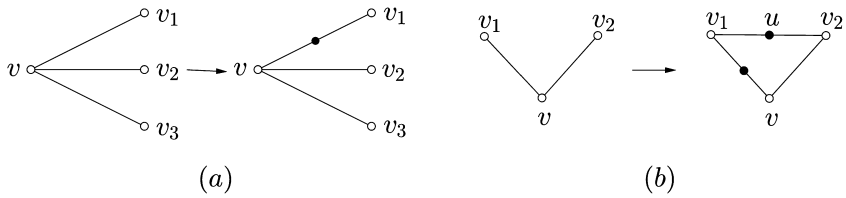

Fig. 2. Illustration of Claim 2.

3. Proof of Theorem 1.3

Suppose to the contrary that G is a counterexample of Theorem 1.3 with minimal μ, i.e., G is a planar graph without adjacent 3-cycles and $\chi_o(G) \geq 8$. According to the minimality of G, for every planar graph H without adjacent 3-cycles, if $\mu(H) < \mu(G)$, then $\chi_o(H) \leq 7$. Our goal is to show that such G does not exist. Embed G in a plane arbitrarily. In order to study the existence of the counterexample G, some forbidden configurations are needed.

3.1. Forbidden configurations

Claim 1. *Any two odd-vertices in G are not adjacent.*

Proof. Suppose by contradiction that u, v are two adjacent odd-vertices in G. Applying $(A1)$ on G by taking $uv := uv$ and $w := w$, we obtain the graph G_1. According to Observation 1, $G_1 \prec G$. By the minimality of G, G_1 admits an odd 7-coloring, denoted by φ_1. Define $\varphi(x) := \varphi_1(x)$ for $x \in V(G)$. Since φ_1 is an odd coloring of G_1, every two adjacent vertices in $G\backslash\{u, v\}$ receive different colors and every vertex in $V(G)\backslash\{u, v\}$ has an odd color, under φ. Since w is a 2-vertex in G_1, $\varphi_1(u) \neq \varphi_1(v)$. So $\varphi(u) \neq \varphi(v)$. We know u, v are odd vertices in G, hence they both have odd colors under φ. Thus φ is an odd 7-coloring of G, which contradicts the assumption that G is not odd 7-colorable. □

Claim 2. *It holds that $\delta(G) \geq 4$.*

Proof. Suppose that there exists a 3^--vertex $v \in V(G)$. Let $N_G(v) = \{v_1\}$ (or $\{v_1, v_2\}$, $\{v_1, v_2, v_3\}$). First assume v is a 1-vertex or a 3-vertex. Applying $(A1)$ on G by taking $uv := vv_1$, we obtain the graph G_1, as shown in Fig. 2(a). By Observation 1, we know $G_1 \prec G$. Hence G_1 admits an odd 7-coloring, denoted by φ_1. Define that $\varphi(x) := \varphi_1(x)$ for $x \in V(G)\backslash\{v\}$ and $\varphi'(v) := c \in [1, 7]\backslash(\varphi(N_G(v)) \cup \varphi_o(N_G(v)))$. Since φ_1 is an odd coloring of G_1, every two adjacent vertices in $G\backslash\{v\}$ receive different colors and every vertex in $V(G)\backslash(\{v\} \cup N_G(v))$ has an odd color, under φ. By the definition of $\varphi'(v)$, $\varphi'(v) \notin \varphi(N_G(v))$. Thus every two adjacent vertices of G receive different colors under $\varphi \cup \varphi'$. Since v is an odd-vertex in G, it has an odd color under $\varphi \cup \varphi'$. For any vertex $x \in N_G(v)$, if it is an odd-vertex, then it certainly has an odd color under $\varphi \cup \varphi'$. Otherwise, x is an even-vertex in G, hence x is an odd-vertex of $G\backslash\{v\}$. So x has an odd color $\varphi_o(x)$ under φ. Since $\varphi'(v) \neq \varphi_o(x)$, $\varphi_o(x)$ is still an odd color of x under $\varphi \cup \varphi'$. Thus every vertex in G has an odd color under $\varphi \cup \varphi'$. Therefore, $\varphi \cup \varphi'$ is an odd 7-coloring of G. This leads to a contradiction.

Assume v is a 2-vertex. We first prove that v_1 and v_2 are 3^+-vertices. Clearly, $d_G(v_i) \neq 1$ for $i \in [1, 2]$. Suppose to the contrary that one of v_1, v_2 is a 2-vertex. Without loss generality, let v_1 be a 2-vertex. Denote by v_{11} the neighbor of v_1 different from v. Applying $(A1)$ on G by taking $uv := vv_1$, we obtain the graph G_1. By Observation 1, $G_1 \prec G$. So G_1 admits an odd 7-coloring, denoted by φ_1. Define that $\varphi(x) := \varphi_1(x)$ for $x \in V(G)\backslash\{v_1, v\}$, $\varphi'(v_1) := c_{v_1} \in [1, 7]\backslash(\varphi(\{v_{11}, v_2\}) \cup \varphi_o(\{v_{11}\}))$ and $\varphi'(v) := c_v \in [1, 7]\backslash(\{c_{v_1}\} \cup \varphi(\{v_2, v_{11}\}) \cup \varphi_o(\{v_2\}))$. Since φ_1 is an odd coloring of G_1, every two adjacent vertices in $G\backslash\{v, v_1\}$ receive different colors and every vertex in $V(G)\backslash\{v, v_1, v_2, v_{11}\}$ has an odd color under φ. According to the definition of φ', we know $\varphi'(v_1) \notin \{\varphi(v_{11}), \varphi'(v)\}$ and $\varphi'(v) \notin \{\varphi'(v_1), \varphi(v_2)\}$. So every two adjacent vertices in G receive different colors. For $x \in \{v_{11}, v_2\}$, if it is an odd-vertex in G, then it has an odd color under $\varphi \cup \varphi'$. Otherwise, x is an even-vertex in G, hence x is an odd-vertex in $G\backslash(N_G(x) \cap \{v_1, v\})$. So x has an odd color $\varphi_o(x)$ under φ. Since $\varphi'(v_1) \neq \varphi_o(v_{11})$ and $\varphi'(v) \neq \varphi_o(v_2)$, $\varphi_o(x)$ is still an odd color of x under $\varphi \cup \varphi'$. Thus every vertex in G has an odd color under $\varphi \cup \varphi'$. So $\varphi \cup \varphi'$ is an odd 7-coloring of G, which contradicts that $\chi_o(G) \geq 8$. Therefore, $d_G(v_i) \geq 3$ for $i \in [1, 2]$.

Applying $(A2)$ on G by taking $xx_1 := vv_1$ and $y_1 := u$, we obtain the graph G_2, as shown in Fig. 2(b). By Observation 1, $G_2 \prec G$. So G_2 admits an odd 7-coloring, denoted by φ_2. Define that $\varphi(x) := \varphi_2(x)$ for $x \in V(G)\backslash\{v\}$ and $\varphi'(v) := c \in [1, 7]\backslash(\varphi(N_G(v)) \cup \varphi_o(N_G(v)))$. Since φ_2 is an odd coloring of G_2, every two adjacent vertices in $G\backslash\{vv_1, vv_2, v_1v_2\}$ receive different colors and every vertex in $V(G)\backslash\{v, v_1, v_2\}$ has an odd color under φ. Since u is a 2-vertex in G_2, $\varphi_2(v_1) \neq \varphi_2(v_2)$. So $\varphi(v_1) \neq \varphi(v_2)$. By the definition of φ', it holds that $\varphi'(v) \notin \varphi(N_G(v))$. Hence every two adjacent vertices in G receive different colors under $\varphi \cup \varphi'$. Since $\varphi(v_1) \neq \varphi(v_2)$, $\varphi(v_1)$ and $\varphi(v_2)$ are odd colors of v under $\varphi \cup \varphi'$. For $x \in \{v_1, v_2\}$, if x is an odd-vertex in G, then it has an odd color under $\varphi \cup \varphi'$. Otherwise, x is an even-vertex in G. Hence x is an odd-vertex in $G\backslash\{v\}$. Consequently, x has an odd color $\varphi_o(x)$ under φ. Since $\varphi'(v) \neq \varphi_o(x)$, $\varphi_o(x)$ is still an odd color of x under $\varphi \cup \varphi'$. So every vertex of G has an odd color under $\varphi \cup \varphi'$. Therefore, φ is an odd 7-coloring of G, which contradicts $\chi_o(G) \geq 8$.

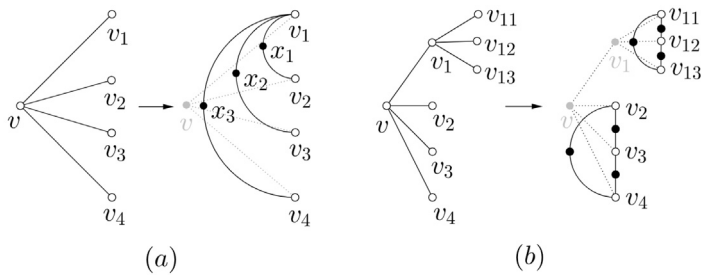

Fig. 3. Illustration of Claim 3.

This proves the claim. □

Claim 3. *Let v be a 4-vertex of G. Then v has at most one odd-neighbor. Furthermore, if v has exactly one odd-neighbor, then v has no 4-neighbors.*

Proof. Assume $N_G(v) = \{v_1, v_2, v_3, v_4\}$. Suppose v has two odd-neighbors, say v_1, v_2, without loss generality. Applying $(A3)$ on G by taking $z := v$, $z_i := v_i$ for $i \in [1, 4]$, $w_i := x_i$ for $i \in [1, 3]$, we obtain the graph G_3, as shown in Fig. 3(a). By Observation 1, we know $G_3 \prec G$. So G_3 admits an odd 7-coloring, denoted by φ_3. Define that $\varphi(x) := \varphi_3(x)$ for $x \in V(G)\backslash\{v\}$ and $\varphi'(v) := c \in [1, 7]\backslash(\varphi(\{v_1, v_2, v_3, v_4\}) \cup \varphi_o(\{v_3, v_4\}))$. Since φ_3 is an odd coloring of G_3, every two adjacent vertices in $G\backslash\{v\}$ receive different colors and every vertex in $G\backslash\{v, v_1, v_2, v_3, v_4\}$ has an odd color under φ. Clearly, it holds that $\varphi'(v) \notin \varphi(N_G(v))$. Hence every two adjacent vertices in $V(G)$ receive different colors under $\varphi \cup \varphi'$. Since x_1, x_2, x_3 are 2-vertices in G_3, $\varphi_3(v_1) \notin \varphi_3(\{v_2, v_3, v_4\})$. Hence $\varphi(v_1) \notin \varphi(\{v_2, v_3, v_4\})$. So $\varphi(v_1)$ is an odd color of v under $\varphi \cup \varphi'$. Since v_1, v_2 both are odd-vertices in G, they both have odd colors under $\varphi \cup \varphi'$. For $x \in \{v_3, v_4\}$, if x is an odd-vertex in G, it certainly has an odd color under $\varphi \cup \varphi'$. Otherwise, x is an even-vertex in G, hence x is an odd-vertex in $G\backslash\{v\}$. It follows that x has an odd color $\varphi_o(x)$ under φ. Since $\varphi'(v) \neq \varphi_o(x)$, $\varphi_o(x)$ is still an odd color of x under $\varphi \cup \varphi'$. So every vertex in $V(G)$ has an odd color under $\varphi \cup \varphi'$. Thus $\varphi \cup \varphi'$ is an odd 7-coloring of G. This contradicts to the assumption that $\chi_o(G) \geq 8$. Therefore, v has at most one odd-neighbor.

Suppose that v has a 4-neighbor and an odd-neighbor. Without loss of generality, assume that v_1 is the 4-neighbor and v_2 is the odd-neighbor of v. Let $N_G(v_1) = \{v, v_{11}, v_{12}, v_{13}\}$. Define $H := G\backslash\{vv_1\}$. Since $d_G(x) \geq 3$ and $d_H(x) \geq 3$ for $x \in \{v, v_1\}$. So $|V_{3^+}(H)| = |V_{3^+}(G)|$ and $|V(H)| = |V(G)|$, it holds that $\mu(H) = \mu(G)$. Applying $(A4)$ on H by taking $z := v, z_1 := v_2, z_2 := v_3$ and $z_3 := v_4$, we obtain the graph H_4. Then applying $(A4)$ again on H_4 by taking $z := v_1$ and $z_i := v_{1i}$ for $i \in [1, 3]$, we obtain the graph $(H_4)_4$, as shown in Fig. 3(b). Let $G^* := (H_4)_4$. By Observation 1, $G^* \prec H$. Together with $\mu(H) = \mu(G)$, we have $G^* \prec G$. Therefore, G^* admits an odd 7-coloring, denoted by φ^*. Define that $\varphi(x) := \varphi^*(x)$ for $x \in V(G)\backslash\{v_1, v\}$, $\varphi'(v_1) := c_{v_1} \in [1, 7]\backslash(\varphi(\{v_{11}, v_{12}, v_{13}\}) \cup \varphi_o(\{v_{11}, v_{12}, v_{13}\}))$, and $\varphi'(v) := c_v \in [1, 7]\backslash(\{c_{v_1}\} \cup \varphi(\{v_2, v_3, v_4\}) \cup \varphi_o(\{v_3, v_4\}))$. Since φ^* is an odd coloring of G^*, every two adjacent vertices in $G\backslash\{v_1, v\}$ receive different colors and every vertex in $V(G)\backslash\{v_1, v, v_{11}, v_{12}, v_{13}, v_2, v_3, v_4\}$ has an odd color under φ. Since $\varphi'(v_1) \notin (\{\varphi'(v)\} \cup \varphi(\{v_{11}, v_{12}, v_{13}\}))$ and $\varphi'(v) \notin (\{\varphi'(v_1)\} \cup \varphi(\{v_2, v_3, v_4\}))$, every two vertices in G receive different colors under $\varphi \cup \varphi'$. Since $|\{\varphi(v_{11}), \varphi(v_{12}), \varphi(v_{13})\}| = 3$ and $d_G(v_1) = 4$, v_1 always has at least two odd colors under $\varphi \cup \varphi'$, no matter what $\varphi'(v)$ is. Similarly, v has at least two odd colors under $\varphi \cup \varphi'$, no matter what $\varphi'(v_1)$ is. For $x \in \{v_{11}, v_{12}, v_{13}, v_2, v_3, v_4\}$, if x is an odd-vertex in G, then it certainly has an odd color under $\varphi \cup \varphi'$. Otherwise, x is an even-vertex in G, hence x is an odd-vertex in $G\backslash\{v, v_1\}$. By the definition of φ', $\varphi_o(x)$ is still an odd color of x under $\varphi \cup \varphi'$. Hence every vertex in G has an odd color under $\varphi \cup \varphi'$. Therefore, $\varphi \cup \varphi'$ is an odd 7-coloring of G, a contradiction.

This proves Claim 3. □

Claim 4. *Let v be a 4-vertex in G and $N_G(v) = \{x, y, z, w\}$. If $xy \in E(G)$ and one of x, y is a 4-vertex, then neither of z and w is a 4-vertex.*

Proof. Without loss of generality, we suppose x and w are 4-vertices. Let $N_G(x) = \{x_1, x_2, v, y\}$ and $N_G(w) = \{v, w_1, w_2, w_3\}$. Define that H is a graph obtained from $G\backslash\{v\}$ by joining z and y with a path zy_1y. Clearly, H is a planar graph without adjacent 3-cycles. The vertices x and w are 3^+-vertices, no matter they are in G or in H. For $u \in \{y, z\}$, since $d_H(u) = |(N_G(u)\backslash\{v\}) \cup \{y_1\}|$, $d_G(u) = d_H(u)$. Together with the fact that $d_G(v) \geq 3$, $|V_{3+}(H)| = |V_{3+}(G)\backslash\{v\}| < |V_{3+}(G)|$ and $|V(H)| = |V(G)|$. It follows that $H \prec G$. For the graph H, applying $(A4)$ on w and its neighbors in H, we delete w and join w_i and w_{i+1} with a path of length two, where the subscript is taken modulo 3. Then we obtain the graph H_4. Applying $(A4)$ again on H_4 by taking $z := x, z_1 := x_1, z_2 := x_2$ and $z_3 := y$, we obtain the graph $(H_4)_4$, as shown in Fig. 4. Let $G^* := (H_4)_4$. By Observation 1, $G^* \prec H$. And we know $H \prec G$, so $G^* \prec G$. Hence G^* admits an odd 7-coloring, denoted by φ^*. Define that $\varphi(u) := \varphi^*(u)$ for $u \in V(G)\backslash\{v, x, w\}$, $\varphi'(x) := c_x \in [1, 7]\backslash(\varphi(\{x_1, x_2, y, z\}) \cup \varphi_o(\{x_1, x_2\}))$,

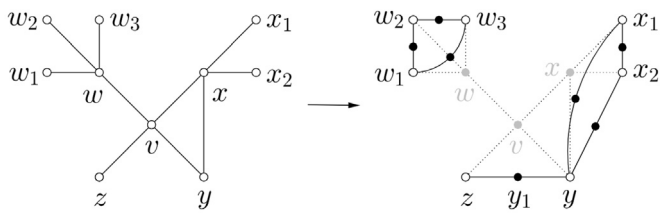

Fig. 4. Illustration of Claim 4.

$\varphi'(w) := c_w \in [1,7]\backslash(\varphi(\{w_1,w_2,w_3\}) \cup \varphi_o(\{w_1,w_2,w_3\}))$ and $\varphi'(v) := c_v \in [1,7]\backslash(\{c_x,c_w\} \cup \varphi(\{z,y\}) \cup \varphi_o(\{z,y\}))$. Since φ^* is an odd coloring of G^*, every two adjacent vertices in $G\backslash\{v,x,w\}$ receive different colors and every vertex in $V(G)\backslash\{v,x,y,z,w,x_1,x_2,w_1,w_2,w_3\}$ has an odd color under $\varphi \cup \varphi'$. For $u \in \{v,x,w\}$, by the definitions of φ and φ', we know $(\varphi \cup \varphi')(u) \notin (\varphi \cup \varphi')(N_G(u))$. Hence every two adjacent vertices in G receive different colors under $\varphi \cup \varphi'$. Since y_1 is a 2-vertex in G^*, $\varphi^*(z) \neq \varphi^*(y)$. So $\varphi(z) \neq \varphi(y)$. By the definitions of φ and φ', $\varphi'(x) \neq \varphi(y)$ and $\varphi'(x) \neq \varphi(z)$. So $|(\varphi \cup \varphi')(N_G(v)\backslash\{w\})| = 3$. Together with the fact that $d_G(v) = 4$, v has at least two odd colors under $\varphi \cup \varphi'$, no matter what $(\varphi \cup \varphi')(w)$ is. Similarly, for $u \in \{x,w\}$, since $|(\varphi \cup \varphi')(N_G(u)\backslash\{v\})| = 3$ and $d_G(u) = 4$, u has at least two odd colors under $\varphi \cup \varphi'$, no matter what $(\varphi \cup \varphi')(v)$ is. For $u \in \{y,z,x_1,x_2,w_1,w_2,w_3\}$, if u is an odd-vertex in G, then it always has an odd color. Otherwise, u is an even-vertex in G, hence u is an odd-vertex in $G\backslash\{v,x,w\}$. So u has an odd color $\varphi_o(u)$ under φ. Since $(\varphi \cup \varphi')(u) \neq \varphi'(N_G(u) \cap \{v,x,w\})$, $\varphi_o(u)$ is still an odd color of u under $\varphi \cup \varphi'$. Thus every vertex in G has an odd color under $\varphi \cup \varphi'$. Therefore, $\varphi \cup \varphi'$ is an odd 7-coloring of G. This leads to a contradiction and this claim is confirmed. □

3.2. Discharging

We use $F(G)$ and $d_G(f)$ to denote the face set of G and the number of edges incident with f in $F(G)$, respectively. Since G is a planar graph, by the Euler's formula

$$|V(G)| - |E(G)| + |F(G)| = 2$$

and Handshaking Lemma

$$\sum_{v \in V(G)} d_G(v) = \sum_{f \in F(G)} d_G(f) = 2|E(G)|,$$

we have

$$\sum_{v \in V(G)} (d_G(v) - 6) + \sum_{f \in F(G)} (2d_G(f) - 6) = -12. \tag{1}$$

If we can prove

$$\sum_{v \in V(G)} (d_G(v) - 6) + \sum_{f \in F(G)} (2d_G(f) - 6) \geq 0, \tag{2}$$

which contradicts the equation (1), then G does not exist. Consequently, Theorem 1.3 is true. Define that

$$\omega_0(v) := d(v) - 6 \quad \text{for} \quad v \in V(G) \quad \text{and}$$
$$\omega_0(f) := 2d(f) - 6 \quad \text{for} \quad f \in F(G).$$

We will design some discharging rules and denote by $\omega(x)$ the final charge of $x \in V(G) \cup F(G)$ after applying the discharging rules. During discharging, for every $x \in V(G) \cup F(G)$, we can guarantee

$$\sum_{x \in V(G) \cup F(G)} \omega(x) = \sum_{x \in V(G) \cup F(G)} \omega_o(x) = \sum_{v \in V(G)} (d_G(v) - 6) + \sum_{f \in F(G)} (2d_G(f) - 6).$$

To prove (2), it suffices to show for every $x \in V(G) \cup F(G)$, we have $\omega(x) \geq 0$. The followings are the discharging rules (Fig. 5).

(R1) Every face $f \in F(G)$ sends $\frac{2d(f)-6}{d(f)}$ to each vertex incident with f.
(R2) For $uv \in E(G)$ with $d_G(u) = 5$ and $d_G(v) = 4$, if uv is incident with a 3-face in G, then u sends $\frac{1}{4}$ to v.
(R3) Let uv be an edge in G with $d_G(u) \geq 6$ and $d_G(v) = 4$.

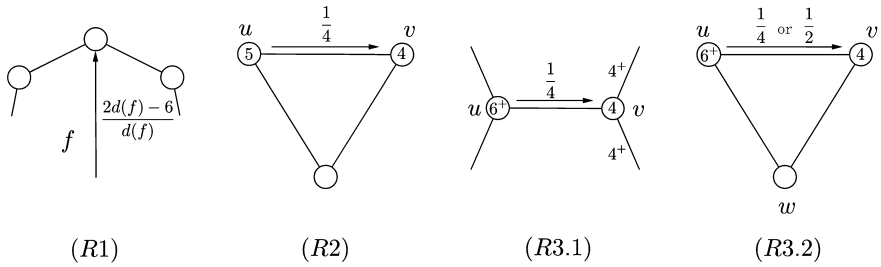

Fig. 5. Discharging rules.

(R3.1) If two faces incident with uv are both 4^+-faces, then u sends $\frac{1}{4}$ to v.

(R3.2) If uv is incident with a 3-face, denoted by uvw, then u sends $\frac{1}{4}$ to v if w is a 4-vertex and $\frac{1}{2}$ otherwise.

Only $(R1)$ changes the charge of faces, it holds that $\omega(f) = \omega_0(f) - (\frac{2d(f)-6}{d(f)}) \times d(f) = 0$ for every $f \in F(G)$, as desired. It remains to show that the final charge of every vertex is non-negative.

Claim 5. *The final charge of every 6^+-vertex in G is non-negative.*

Proof. Let v be a 6^+-vertex in G and k be the number of 4^+-faces incident with v. Since any two 3-cycles of G are not adjacent, $\lceil \frac{d(v)}{2} \rceil \le k \le d(v)$. The vertex v receives at least $\frac{1}{2}k$ from its incident 4^+-faces after applying $(R1)$ and sends at most $\frac{1}{2}(d(v)-k)$ to its adjacent vertices after applying $(R3.2)$. Notice that the vertex v also sends charge to its adjacent vertices by applying $(R3.1)$. We know there are k many 4^+-faces incident with v, hence there are at most k vertices in $N_G(v)$ that each receives $\frac{1}{4}$ from v by applying $(R3.1)$. Above all, $\omega(v) \ge \omega_0(v) + \frac{1}{2}k - \frac{1}{2}(d(v)-k) - \frac{1}{4}k = \frac{1}{2}d(v) + \frac{3}{4}k - 6$. If $d(v) \ge 7$, since $k \ge \lceil \frac{d(v)}{2} \rceil \ge \frac{d(v)}{2}$, $\omega(v) \ge \frac{1}{2}d(v) + \frac{3}{4} \times \frac{1}{2}d(v) - 6 \ge \frac{7}{8}d(v) - 6 \ge \omega(v) \ge \frac{7}{8} \times 7 - 6 = \frac{1}{8}$. Assume $d(v) = 6$. If $k \ge 4$, then $\omega(v) \ge \frac{1}{2} \times 6 + \frac{3}{4} \times 4 - 6 = 0$. Assume $k = 3$. In this case, since any two 4^+-faces incident with v are not adjacent, v can not send charge to its neighbors by applying $(R3.1)$. So $\omega(v) = \omega_0(v) + \frac{3}{2} - \frac{3}{2} = 0 + \frac{3}{2} - \frac{3}{2} = 0$. □

Claim 6. *The final charge of every 5-vertex in G is non-negative.*

Proof. Let v be a 5-vertex in G. Since any two 3-faces are not adjacent in G, there are at least three 4^+-faces incident with v. By $(R1)$, v receives at least $\frac{1}{2} \times 3 = \frac{3}{2}$ from its incident faces. According to Claim 3, every 3-face incident with v has at most one incident 4-vertex. By $(R2)$, v sends at most $\frac{1}{4} \times 2 = \frac{1}{2}$ to its adjacent vertices. Thus $\omega(v) \ge \omega_0(v) + \frac{1}{2} \times 3 - \frac{1}{4} \times 2 = (5-6) + \frac{1}{2} \times 3 - \frac{1}{4} \times 2 = 0$. □

Claim 7. *The final charge of every 4-vertex in G is non-negative.*

Proof. Let v be a 4-vertex in G and $N_G(v) = \{v_1, v_2, v_3, v_4\}$. Denote by f_{12} the face incident with vertices v_1, v, v_2. Similarly, we have f_{23}, f_{34}, f_{41}. We know every 4^+-face sends at least $\frac{1}{2}$ to each its incident vertex, by applying $(R1)$. If all faces incident with v are 4^+-faces, then $\omega(v) \ge \omega_o(v) + \frac{1}{2} \times 4 = (4-6) + \frac{1}{2} \times 4 = 0$.

Assume that there exists exactly one 3-face incident with v, without loss of generality, denoted by f_{12}. If v_1, v_2 are both 5^+-vertices, by Claim 3, there is at most one 5-vertex in $\{v_1, v_2\}$. So v receives at least $\frac{1}{2} \times 3 = \frac{3}{2}$ from its incident faces by $(R1)$ and receives at least $\frac{1}{4} + \frac{1}{2} = \frac{3}{4}$ from v_1 and v_2, by $(R2)$ and $(R3.2)$. Thus $\omega(v) \ge \omega_0(v) + \frac{3}{2} + \frac{3}{4} = (4-6) + \frac{3}{2} + \frac{3}{4} = \frac{1}{4}$. If one of v_1, v_2 is a 4-vertex, then v_3, v_4 are both 6^+-vertices by Claim 3 and Claim 4. Since f_{23}, f_{34}, f_{41} are 4^+-faces, according to $(R3.1)$, v_3 and v_4 send $\frac{1}{4} \times 2 = \frac{1}{2}$ to v. By $(R1)$, each 4^+-face incident with v sends at least $\frac{1}{2}$ to v. So $\omega(v) \ge \omega_0(v) + \frac{1}{2} \times 3 + \frac{1}{4} \times 2 = (4-6) + \frac{1}{2} \times 3 + \frac{1}{4} \times 2 = 0$.

Assume there exist exactly two 3-faces incident with v, denoted by f_{12} and f_{34}. Assume $d(v_i) \ge 5$ for every $i \in [1, 4]$. By Claim 3, at most one vertex in $\{x_1, x_2, x_3, x_4\}$ is 5-vertex. In this case, v receives at least $\frac{1}{2} \times 2 = 1$ from f_{23} and f_{41} by $(R1)$ and receives at least $\frac{1}{4} + \frac{1}{2} \times 3 = \frac{7}{4}$ from its adjacent vertices by $(R2)$ and $(R3.2)$. Thus $\omega(v) \ge \omega_0(v) + 1 + \frac{7}{4} = (4-6) + 1 + \frac{7}{4} = \frac{3}{4}$. If there exists a 4-vertex in $\{x_1, x_2, x_3, x_4\}$, denoted by v_1, then v_3 and v_4 are both 6^+-vertices by Claim 3 and Claim 4. The vertex v receives at least $\frac{1}{2} \times 2 = 1$ from its f_{23} and f_{41} by $(R1)$ and receives $\frac{1}{2} \times 2 = 1$ from its neighbors v_3 and v_4 by $(R3.2)$. Thus $\omega(v) \ge \omega_0(v) + 1 + 1 \ge (4-6) + 1 + 1 = 0$.

Therefore, the final charge of every 4-vertex is non-negative.

To summarize Claim 5, Claim 6 and Claim 7, the final charge of every vertex in G is non-negative. This completes the proof of Theorem 1.3. □

4. Proof of Theorem 1.4

Similar to the proof of Theorem 1.3, suppose G is a counterexample of Theorem 1.4 with minimal μ, i.e., G is a planar graph without 3-cycles and intersecting 4-cycles and $\chi_o(G) \geq 6$. According to the minimality of G, for any planar graph H without 3-cycles and intersecting 4-cycles, if $\mu(H) < \mu(G)$, then H admits an odd 5-coloring. Embed G in a plane arbitrarily. Some configurations are forbidden by claims in Subsection 4.1.

4.1. Forbidden configurations

Claim 8. *It holds that $\delta(G) \geq 3$.*

Proof. Suppose by contradiction that v is a 2^--vertex in G. First assume v is a 1-vertex in G. Let v_1 be the neighbor of v. Applying $(A1)$ on G by taking $uv := vv_1$ and $w := w$, we obtain the graph G_1. By Observation 2, $G_1 \prec G$. Hence G_1 admits an odd 5-coloring, denoted by φ_1. Define that $\varphi(x) := \varphi_1(x)$ for $x \in V(G)\backslash\{v\}$ and $\varphi(v) := \varphi_1(w)$. Since φ_1 is an odd coloring of G_1, every two adjacent vertices in $G\backslash\{v\}$ receive different colors and every vertex in $V(G)\backslash\{v, v_1\}$ has an odd color under φ. Since $wv_1 \in E(G_1)$, $\varphi_1(w) \neq \varphi_1(v_1)$. So $\varphi(v) \neq \varphi(v_1)$, it follows that every two adjacent vertices receive different colors under φ. Since v is a 1-vertex in G, the color on its neighbor is always the odd color of it. Since φ_1 is an odd coloring of G_1, there exists a color c appearing odd times on $N_{G_1}(v_1)$ under φ_1. Since $N_G(v_1)\backslash\{v\} = N_{G_1}(v_1)\backslash\{w\}$ and $\varphi(v) = \varphi_1(w)$, c is still an odd color of v_1 under φ. So every vertex in G has an odd color. Therefore, φ is an odd 5-coloring of G. This leads to a contradiction.

Assume v is a 2-vertex and $N_G(v) = \{v_1, v_2\}$. We first show that all neighbors of v are 3^+-vertices. Clearly, v_1, v_2 are 2^+-vertices. Suppose v has a 2-neighbor in G, say v_1, without loss of generality. Let $N_G(v_1) = \{v, v_{11}\}$. Applying $(A1)$ on G by taking $uv := vv_1$, we obtain the graph G_1. By Observation 2, $G_1 \prec G$. So G_1 admits an odd 5-coloring, denoted by φ_1. Define that $\varphi(x) := \varphi_1(x)$ for $x \in V(G)\backslash\{v_1, v\}$, $\varphi'(v_1) := c_{v_1} \in [1, 5]\backslash(\varphi(\{v_{11}, v_2\}) \cup \varphi_o(\{v_{11}\}))$ and $\varphi'(v) := c_v \in [1, 5]\backslash\{c_{v_1}, \varphi(v_{11}), \varphi(v_2), \varphi_o(v_2)\}$. Similar to the proof of Claim 2, we can verify that $\varphi \cup \varphi'$ is an odd 5-coloring of G. This contradicts that G is not odd 5-colorable. Thus $d_G(v_i) \geq 3$ for $i \in [1, 2]$.

Applying $(A2)$ on G by taking $x := v, x_1 := v_1, x_2 := v_2$ and $y_1 := u$, we obtain the graph G_2. By Observation 2, $G_2 \prec G$. Hence G_2 has an odd 5-coloring, denoted by φ_2. Define that $\varphi(x) := \varphi_2(x)$ for $x \in V(G)\backslash\{v\}$ and $\varphi'(v) := c \in [1, 5]\backslash(\varphi(\{v_1, v_2\}) \cup \varphi_o(\{v_1, v_2\}))$. We can verify $\varphi \cup \varphi'$ is an odd 5-coloring of G by imitating the proof of Claim 2. This contradicts that $\chi_o(G) \geq 6$.

Therefore, $\delta(G) \geq 3$. □

Claim 9. *Any two odd vertices in G are not adjacent.*

The proof of Claim 9 is analogous to the one of Claim 1, so we omit the proof here.

4.2. Discharging

By the Euler's formula and Handshaking lemma, we have

$$\sum_{v \in V(G)} (2d_G(v) - 6) + \sum_{f \in F(G)} (d_G(f) - 6) = -12. \tag{3}$$

Similar to the proof of Theorem 1.3, if we can prove

$$\sum_{v \in V(G)} (2d_G(v) - 6) + \sum_{f \in F(G)} (d_G(f) - 6) \geq 0, \tag{4}$$

which contradicts the equation (3), then Theorem 1.4 follows immediately. Define that

$$\omega_0(v) := 2d(v) - 6 \quad \text{for} \quad v \in V(G) \quad \text{and}$$
$$\omega_0(f) := d(f) - 6 \quad \text{for} \quad f \in F(G).$$

We will design some discharging rules and denote by $\omega(x)$ the final charge of $x \in V(G) \cup F(G)$ after applying the discharging rules. During discharging, for every $x \in V(G) \cup F(G)$, we guarantee

$$\sum_{x \in V(G) \cup F(G)} \omega(x) = \sum_{x \in V(G) \cup F(G)} \omega_o(x) = \sum_{v \in V(G)} (2d_G(v) - 6) + \sum_{f \in F(G)} (d_G(f) - 6).$$

To prove (4), it suffices to show for every $x \in V(G) \cup F(G)$, we have $\omega(x) \geq 0$. The follows are the discharging rules.

(R1) Every 4^+-vertex sends $\frac{1}{2}$ to each its incident face.

(R2) Every 5^+-face sends $\frac{1}{4}$ to each its adjacent 4-face.

Claim 10. *The final charge of every vertex in G is non-negative.*

Proof. Clearly, if v is a 3-vertex in G, then $\omega(v) = \omega_0(v) = 2 \times 3 - 6 = 0$. Assume v is a 4^+-vertex in G. By $(R1)$, $\omega(v) = \omega_0(v) - \frac{1}{2} \times d(v) = (2d(v) - 6) - \frac{1}{2} \times d(v) = \frac{3}{2}d(v) - 6 \geq 0$. So $\omega(v) \geq 0$ for every $v \in V(G)$. □

Claim 11. *The final charge of every 4-face in G is non-negative.*

Proof. Let f be a 4-face in G. Since any two odd-vertices in G are not adjacent, there are at most two 3-vertices incident with f. So f receives at least $\frac{1}{2} \times 2 = 1$ from its incident vertices by $(R1)$. Since any two 4-faces in G do not intersect, all faces adjacent to f are 5^+-faces. So f receives at least $\frac{1}{4} \times 4$ from its adjacent faces by $(R2)$. Thus $\omega(v) \geq \omega_0(v) + 1 + 1 = (4 - 6) + 1 + 1 = 0$. □

Claim 12. *The final charge of every 5^+-face in G is non-negative.*

Proof. Let f be a 5^+-face in G. Since any two odd-vertices in G are not adjacent, there are at most $\lfloor \frac{1}{2}d(f) \rfloor$ many 3-vertices incident with f. Since any two 4-faces do not intersect, there are at most $\lfloor \frac{1}{2}d(f) \rfloor$ many 4-faces adjacent to f. Therefore, after applying $(R1)$ and $(R2)$, $\omega(f) \geq \omega_0(f) + \frac{1}{2} \times \lceil \frac{1}{2}d(f) \rceil - \frac{1}{4} \times \lfloor \frac{1}{2}d(f) \rfloor \geq 0$. □

The proof of Theorem 1.4 is completed.

5. Proof of Theorem 1.5

Let $H = T \cup C_n$ be a Halin graph, where C_n is a cycle with n vertices. In the following, we prove H is odd 4-colorable. Assume first that H is a wheel graph, i.e., T is a star with at least four vertices. Let $C_n = u_1u_2 \ldots u_nu_1$ and u_0 be the central vertex of T. Define

$$\varphi(u_i) := \begin{cases} 1, & \text{if} \quad i = 0; \\ 2, & \text{if} \quad i = n; \\ 3, & \text{if} \quad i \in [1, n-1] \quad \textit{and} \quad i \equiv 1 \pmod 2; \\ 4, & \text{if} \quad i \in [1, n-1] \quad \textit{and} \quad i \equiv 0 \pmod 2. \end{cases}$$

Obviously, $\varphi(x) \neq \varphi(y)$ for every $xy \in E(H)$. Since the color 2 appears exactly once on $N_H(u_0)$ and the color 1 appears exactly once on $N_H(u_i)$ for every $i \in [1, n]$, every vertex in H has an odd color. So φ is an odd 4-coloring of H. Note that $\chi_o(H) = 4$ when $n \equiv 0 \pmod 4$, and $\chi_o(H) = 3$ when $n \equiv 2 \pmod 4$. Hence the upper and lower bounds in Theorem 1.5 are tight.

Assume next H is not a wheel graph. In this case, $|V(H \backslash V(C_n))| \geq 2$. Let $T' := H \backslash V(C_n)$. We choose an arbitrarily vertex $x_1 \in V(T')$ as the root of T' and T. Given two vertices x, y in H, we use $d_H(x, y)$ to denote the number of edges in the shortest path between x and y in H. For any two adjacent vertices x, y in T, if $d_T(x_1, x) < d_T(x_1, y)$, then x is called the *father* of y and y is called the *child* of x. The father of y is denoted by $f(y)$. Given a vertex set $Y \subseteq V(H)$, if all vertices in Y have a common father, we denote it by $f(Y)$.

Claim 13. *Let x be a vertex from $V(T') \backslash (\{x_1\} \cup N_{T'}(x_1))$. Then there exists an odd 3-coloring φ of T' such that $\varphi(x)$ is different from $\varphi(f(x))$ and $\varphi(f(f(x)))$.*

Proof. Let $N_{T'}(x_1) = \{x_{11}, x_{12}, \ldots, x_{1d_{T'}(x_1)}\}$. Firstly, define $\varphi(x_1) := 1$, $\varphi(x_{1i}) := 2$ for $i \in [1, d_{T'}(x_1) - 1]$, and $\varphi(x_{1d_{T'}(x_1)}) := 3$. Next, we color the vertices in $V(T') \backslash (\{x_1\} \cup N_{T'}(x_1))$. For every $x \in V(T') \backslash (\{x_1\} \cup N_{T'}(x_1))$, if $f(x)$ and $f(f(x))$ have been colored, then define $\varphi(x) := c \in [1, 3] \backslash \{\varphi(f(x)), \varphi(f(f(x)))\}$. Keeping doing this, we obtain a coloring φ of $V(T')$ such that

$$\varphi(x) \notin \{\varphi(f(x)), \varphi(f(f(x)))\} \quad \text{for any} \quad x \in V(T') \backslash (\{x_1\} \cup N_{T'}(x_1)). \tag{5}$$

Clearly, φ is a proper coloring of T'. It suffices to show that every vertex of T' has an odd color. Let $X = \{v \mid f(v) = x\}$. According to (5), it holds that $\varphi(f(x)) \notin \varphi(X)$. Since $N_{T'}(x) = X \cup \{f(x)\}$, it holds that every $x \in V(T') \backslash \{x_1\}$ has an odd color, which is $\varphi(f(x))$. Note that the color 3 appears exactly once on $N_{T'}(x_1)$, so x_1 has an odd color. Therefore, φ is an odd 3-coloring of T'. □

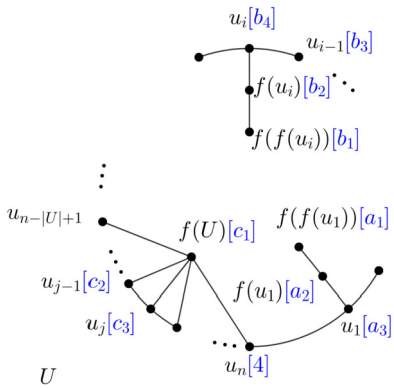

Fig. 6. Illustration of the grouping and coloring of C_n: (1) for vertex $x \in V(H)$ and color $d \in \{a_1, a_2, a_3, b_1, b_2, b_3, b_4, c_1, c_2, c_3, 4\}$, $x[d]$ indicates that d is the color received by x; (2) $a_1, a_2, a_3 \in [1, 3]$ and $a_3 \notin \{a_1, a_2\}$, $b_1, b_2, b_3, b_4 \in [1, 4]$ and $b_4 \notin \{b_1, b_2, b_3\}$, $c_1, c_2, c_3 \in [1, 3]$ and $c_3 \notin \{c_1, c_2\}$; (3) $i \in [2, n - |U|]$ and $j \in [n - |U| + 1, n - 1]$.

The vertex set $V \subset V(C_n)$ is said to be a *group*, if it is the largest set that meets the following two conditions: (1) all vertices in V have a common father in T; (2) The graph induced by V is a path or a vertex. Halin graphs are planar graphs and all vertices of Halin graphs are 3^+-vertices, so there exists a group, denoted by U, whose size is at least 2. Denote the first vertex in $V(C_n)\backslash U$ by u_1, in counterclockwise order. Starting from u_1 and moving counterclockwise, denote the vertices in C_n by $u_1, u_2, \ldots, u_n$, respectively. Let φ be an odd coloring of T' satisfying the condition of Claim 13. For $u_i \in V(C_n)$, define

$$\alpha(u_i) := \varphi(f(u_i)) \quad \text{and} \quad \beta(u_i) := \begin{cases} \varphi(f(f(u_i))), & \text{if} \quad f(u_i) \neq x_1; \\ 3 \quad , & \text{if} \quad f(u_i) = x_1. \end{cases}$$

By coloring $u_1, u_2, \ldots, u_n$ in order, we will give C_n a proper coloring ϕ such that $\varphi \cup \phi$ is an odd 4-coloring of H. First color u_1 with $\phi(u_1) \in [1, 3]\backslash\{\alpha(u_1), \beta(u_1)\}$. Assume now that for every $i \in [2, n-1]$, $u_1, \ldots, u_{i-1}$ have been colored. We next color u_i with $\phi(u_i) \in [1, 4]\backslash\{\alpha(u_i), \beta(u_i), \phi(u_{i-1})\}$ if $i \in [2, n - |U|]$ and color u_i with $\phi(u_i) \in [1, 3]\backslash\{\alpha(u_i), \phi(u_{i-1})\}$ otherwise. Finally, color u_n with $\phi(u_n) = 4$ (Fig. 6).

Clearly, under $\varphi \cup \phi$, every two adjacent vertices in H receive different colors. For $x \in V(H)\backslash(\{x_1, f(U)\})$, $\varphi(f(x))$ is the odd color of x. Note that the color 3 $(= \varphi(x_{1d_{T'}(x_1)}))$ is an odd color of x_1 and the color 4 $(= \phi(u_n))$ is an odd color of $f(U)$. Thus $\varphi \cup \phi$ is an odd 4-coloring of H.

Therefore, every Halin graph admits an odd 4-coloring. □

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

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